HSI-FrFRWKV: Fractional Fourier RWKV for Cross-Scene Coastal Wetland Mapping on Hyperspectral Images

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Abstract*—Coastal wetlands have irreplaceable value in global ecological balance and biodiversity conservation, and hyperspectral images (HSIs) provide support for their fine mapping and protection measure formulation. Due to the dynamic changes of wetlands, existing cross-domain classification methods usually ignore frequency and phase information of HSIs and struggle to cope with domain shifts caused by sensors, lighting, and phase changes, and Transformer-based methods usually have quadratic complexity. As such, a receptance-weighted key value (RWKV) with linear complexity and parallel processing advantages is introduced to propose the first fractional Fourier transform (FrFT) RWKV model for cross-domain wetland mapping of HSIs, namely HSI-FrFRWKV. First, an FrFT-RWKV block is designed, with heterogeneous token shift and FrFT-WKV attention mechanisms as its cores for multiscale local context and global dependency learning. A hierarchical FrFT-RWKV network (FRNet) is then built to extract amplitude–phase-modulated spatial–spectral features that have not been fully explored. Moreover, to better focus on frequency and structure information of HSIs, a parameter-free amplitude–phase learning method in the optimal domain between the space and frequency domains (FAP loss) is designed as a consistency constraint, ensuring clarity and structure of predictions in FRNet. Finally, a bi-classifier adversarial network and a replay-based self-distillation learning method are devised to improve training stability for cross-domain wetland mapping. A public cross-domain classification dataset and two wetland mapping datasets are adopted, whose experimental results demonstrate that the proposed HSI-FrFRWKV outperforms other state-of-the-art methods. Our source code will be available at https://github.com/WenshuaiHu/HSI-FrFRWKV

***Index Terms*—Amplitude–phase analysis, cross-scene wetland mapping, fractional Fourier transform (FrFT), hyperspectral image (HSI), receptance weighted key value (RWKV), self-distillation learning.**

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I. Introduction

COASTAL wetlands, one of the ecosystems with notable ecological value on Earth, provide habitats for animals and plants, human food, and industrial raw materials. They play a key role in climate and hydrological regulation, promoting carbon sequestration and emission reduction and maintaining ecological balance [1], [2]. Reliable monitoring methods are crucial for their restoration, protection, and sustainable management. The development of remote sensing technology has provided a new way for wetland monitoring [3], [4].

Hyperspectral remote sensing allows hyperspectral images (HSIs) with spatial details and spectral information [5], [6] to be an effective means of fine monitoring of coastal wetlands [7]. In recent years, deep learning has gradually become a key driving force in the field of remote sensing [8], [9], [10] and been a key solution for wetland mapping on HSIs [11], with various deep neural networks emerging [12], [13], [14]. However, in practical applications, there are still some challenges. First, the dynamic characteristics of wetlands (i.e., tidal fluctuations and phenological changes) result in significant spectral variations over time and environment. Second, wetlands have a variety of land-cover types and mixed pixels, and collecting samples is costly and time-consuming due to remote and complex environments. In addition, HSIs are usually collected in different geographical regions, at different times, or using different sensors. Wetland types and spectral characteristics are completely different, resulting in serious domain shift between scenes, making it difficult for the above models trained on the source domain to transfer to the target domain, limiting their practical applications [1], [15].

Domain adaptation (DA), as one of the transfer learning methods, has been proved to be an effective method to overcome the above problems [16]. According to differences in implementation methods and principles, existing cross-domain HSI classification methods can be divided into the feature-, instance-, and classifier-based methods [2]. The feature-based one aims to learn domain-invariant representations by minimizing distribution differences between domains, with maximum mean discrepancy (MMD) and adversarial DA being two typical representatives. The instance-based one allocates appropriate weights to reuse source samples based on predefined rules to align with the target distribution [17], [18]. The classifier-based one adjusts its parameters trained in the source domain to adapt to the target domain. Among

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them, the feature- and classifier-based methods were the most commonly used.

For the feature-based method, a multilevel unsupervised DA framework was designed by integrating local MMD (LMMD) and contrastive learning, achieving image-, feature-, and logic-level alignments for cross-scene HSI classification [19]. A shift-sensitive spatial–spectral disentangling learning (S^4DL) approach was designed, with gradient-guided spatial–spectral decomposition and a shift-sensitive adaptive monitor [20]. Based on latent adversarial disentanglement and instance shuffling, an adversarial decoupling domain generalization network (ADNet) was built [21]. A high-resolution DA network was devised for feature- and class-level alignments [22]. Hu et al. [23] proposed a dual-adversarial network with multichannel soft label learning and entropy constraint for class-wise semantic and structural alignments. He further designed an extended multimodal long short-term memory (xMLSTM) [24], forming a cross-modal compensation-consistent U-shape masked xMLSTM neural network to address imbalanced representation and missing intermodal associations during cross-domain generation, thus improving the cross-domain joint classification performance. For the classifier-based method, Zhou and Ma [25] proposed a cross-domain extreme learning machine to achieve DA by minimizing classification loss and aligning domain distributions with MMD. A joint classifier adaptation and adversarial learning method was devised by using a dual-branch network and dual classifiers, extracting invariant spatial features and reducing spectral discrepancies [2]. Fang et al. [26] built a two-stage DA framework, with self-distillation and test-time adaptation to promote training stability and refine the bi-classifier. With LMMD and self-training, a dual classification head self-training network (DHSNet) was built to reduce domain shift and pseudo-label noise [27].

Although the above methods have achieved good performance, they ignore the utilization of frequency information, resulting in incomplete feature extraction of land covers and cross-coupling of semantic information [28]. Most existing HSI processing methods were based on the wavelet transform (WT) [29] or Fourier transform (FT) [30], [31]; however, these global transforms were more suitable for stationary signals, which will bring limitations when applied to nonstationary signals such as HSIs [32]. As such, the FT was extended to the fractional FT (FrFT), with the ability to describe frequency characteristics of signals in local space, making it suitable for nonstationary HSIs [33]. The FrFT can enable the model to capture clearer class-wise boundary information and alleviate the phenomenon of semantic cross coupling [34]. Therefore, some novel FrFT-based methods have been proposed for HSIs, such as a fractional domain information-based cross-domain few-shot learning method [28], dynamic multiple fractional Fourier domains fusion [35], and Chirplet Fourier analysis network [36]. However, existing frequency analysis methods often ignored the phase features that characterize the texture and structure of HSIs, resulting in the loss of details during reconstruction and domain transfer, limiting their performance in complex scenes [36].

Unlike existing deep learning components (i.e., convolution, Transformer, and LSTM), a receptance-weighted key value (RWKV) [42] combines global modeling capability and parallel training advantage with efficient inference characteristics, reducing computational complexity while maintaining competitive accuracy. These advantages make it well-suited for cross-scene wetland mapping tasks that involve computational resource constraints or large-scale data processing.

Inspired by the above work and the above advantages of RWKV, a new FrFT RWKV model is proposed for cross-domain wetland mapping on HSIs, namely HSI-FrFRWKV. First, the FrFT is introduced to design an FrFT-based RWKV (FrFT-RWKV) block, forming a hierarchical FrFT-RWKV network (FRNet) to extract amplitude–phase-modulated spatial–spectral features. To improve the clarity of predictions and explore the actual structure information, an amplitude–phase loss in the FrFT domain (FAP) is designed as the optimization objective of FRNet. Finally, a bi-classifier adversarial network and a replay-based self-distillation learning method are devised for cross-domain alignment and training instability elimination, yielding more domain-invariant spatial–spectral features for better cross-domain wetland mapping. The main contributions lie in the following four aspects.

1) A new FrFT-RWKV block is designed through periodic amplitude and phase learning, enhancing the learning of texture, detail, and structural information of HSIs and enlarging the receptive field to adapt to the complex spatial distribution of land covers in coastal wetlands.
2) A hierarchical FRNet is built, with a parameter-free FAP loss designed as its optimization function, serving as a more effective alternative to L2 loss. Therefore, more accurate data generation and discriminative amplitude–phase-modulated spatial–spectral representation can be achieved in the optimal domain between the space and frequency domains.
3) An effective replay-based self-distillation learning method is proposed to alleviate the catastrophic forgetting and unstable training of the proposed HSI-FrFRWKV model, improving cross-scene wetland mapping performance of HSIs.
4) Extensive experiments on a public dataset and two wetland mapping datasets verify that the proposed model can be superior to other state-of-the-art methods.

The rest of this article is organized as follows. Section II introduces the related work. Section III describes the proposed HSI-FrFRWKV in detail. Experimental results and analysis are presented in Section IV. Section V summarizes this article.

II. Related Work

A. Cross-Scene Wetland Mapping

The highly dynamic nature and the subtle differences between similar materials in coastal wetlands make cross-scene wetland mapping a challenging problem. Early wetland mapping only focused on the study of conventional HSI classification methods [11], [12], [13], [14]. Subsequently, various DA methods emerged, improving the generalization and practical application effectiveness of the above methods. Huang et al. [1] built the first cross-scene wetland mapping

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dataset around the Yellow River Estuary and Yancheng wetlands and proposed a spatial–spectral weighted adversarial DA (SSWADA) network. Xin et al. [15] also introduced a cross-scene wetland mapping dataset around the Yellow River Delta and designed a feature disentanglement-based DA network (FDDAN) for class-specific domain-invariant feature learning. A joint classifier adaptation and adversarial learning method was proposed, achieving semantic enhancement in a lightweight manner and reducing training costs while improving interpretability [2]. With the collaboration of HSIs and light detection and ranging (LiDAR), an instance-wise domain generalization network was built by introducing a discrete cosine transform and Gaussian filter-based extended domain for cross-scene wetland mapping [37].

Unlike the above methods, we approach the cross-scene wetland mapping task from the perspective of frequency-domain analysis. Specifically, the amplitude and phase analysis of FrFT is integrated into the RWKV block to design a new FrFRWKV block, coupled with a parameter-free amplitude–phase loss to fully explore the texture, detail, and structural information of HSIs with a large receptive field. Hence, more representative spatial–spectral–frequency features can be carefully extracted to adapt to the complex distribution of land covers for cross-scene wetland mapping on HSIs.

B. Frequency Learning and the FrFT

1) Definition of the FrFT: By rotating the signal in the time–frequency plane, the classical FT is generalized to the FrFT [38].

Given a function $f(t)$, its α-order FrFT can be written as

$$\mathcal{F}^{\alpha}[f(t)](u) = \int_{-\infty}^{\infty} K_{\alpha}(t,u) f(t)\, dt \tag{1}$$

where t means the time or spatial coordinate, u is the coordinate in the fractional domain, and $\alpha \in \mathbb{R}$ denotes the fractional order. The kernel $K_{\alpha}(t,u)$ is defined as

$$K_{\alpha}(t,u) = \begin{cases} \gamma & \text{if } \phi \not\equiv 0 \pmod{\pi} \\ \delta(t-u) & \text{if } \phi \equiv 0 \pmod{2\pi} \\ \delta(t+u) & \text{if } \phi \equiv \pi \pmod{2\pi} \end{cases} \tag{2}$$

where $\gamma = A_{\phi} \exp(i\pi(t^2+u^2)\cot\phi - i2\pi tu \csc\phi)$ and $\phi = \alpha(\pi/2)$ is the rotation angle. The amplitude coefficient A_{ϕ} is obtained by

$$A_{\phi} = \sqrt{\frac{1-i\cot\phi}{2\pi}}. \tag{3}$$

This kernel forms an orthonormal basis of Chirplet functions.

2) Properties of the FrFT: Next, we will derive how the parameter α of FrFT affects the resolution of spatial–spectral analysis for HSIs by analyzing the Wigner distribution (WD) [39].

For a given signal $f(t)$, its WD that represents energy density in the time–frequency plane is described as

$$W_f(t,\omega) = \int_{-\infty}^{\infty} f\left(t+\frac{\tau}{2}\right) f^{*}\left(t-\frac{\tau}{2}\right) e^{-i\omega\tau}\, d\tau \tag{4}$$

where $*$ is the complex conjugate. Let $f_{\alpha}(u)$ be the FrFT of $f(t)$ with angle ϕ. The relationship between WD ($W_{f_{\alpha}}$) of the transformed signal and the original signal is a coordinate rotation

$$W_{f_{\alpha}}(u,v) = W_f(t\cos\phi - \omega\sin\phi, t\sin\phi + \omega\cos\phi). \tag{5}$$

HSIs contain the nonstationary features (i.e., texture gradients), acting as the linear chirps in signal processing [36]. The form of a linear chirp is e^{ikt^2}, and in the standard time–frequency plane (t,ω), its energy is concentrated along a line $\omega = 2kt$, and k is the approximate chirp rate. With the rotation angle ϕ to align the axis u perpendicular to this line, the projection of energy onto it will become a delta function δ

$$\mathcal{F}^{\alpha}(u) \approx \delta(u-u_0). \tag{6}$$

Based on the above analysis, by tuning the fractional order α, the model can effectively separate the features that overlap in both the spatial domain ($\alpha = 0$) and the frequency domain ($\alpha = 1$).

III. Proposed HSI-FrFRWKV Model

A. Architecture Overview

Fig. 1 shows the overall architecture of the proposed HSI-FrFRWKV model for cross-scene wetland mapping on HSIs. Suppose that the HSI in the source and target domains are $\mathcal{S} = \{\mathcal{X}^s, Y^s\}$ and $\mathcal{T} = \{\mathcal{X}^t, Y^t\}$, where s and t mean the source and target domains, respectively. Y is the label space with N classes, where Y^t is unknown in the training stage. The pixels in a local window with size $s \times s$ are extracted for each pixel x, forming 3-D volumes X^s and $X^t \in \mathbb{R}^{s\times s\times C}$ as the input of the whole model. C is the number of spectral bands.

In the feature extraction stage, X^s and X^t are fed into the stem block based on the 2-D convolutional layer to align their channel dimensions, whose outputs undergo an FRNet (denoted as the generator G), as shown in part I of Fig. 1. For its encoder, l FrFT-RWKV blocks (refer to Section III-B) and a downsampling block are alternately used, followed by another FrFT-RWKV block to connect the decoder and extract amplitude–phase-modulated spatial–spectral features, denoted as F^s and F^t. The downsampling block uses a 3×3 depth-wise convolution with stride 2 to reduce their spatial dimension by half, followed by a 1×1 convolution with stride 1 to double the channel number. For its decoder, an upsampling block, a fusion projector, and an FrFT-RWKV block are utilized to reconstruct HSI data, represented as $\bar{X}^s$ and $\bar{X}^t$. The upsampling block contains an upsampling layer with bilinear interpolation and a 1×1 convolution, doubling the spatial size and halving the channel number. The fusion projector contains the concatenation operation and a 1×1 convolution for coarse-to-fine alignment. To enable the designed FRNet to better explore frequency and structure information of HSIs, inspired by [40], a new amplitude–phase loss in the FrFT domain (FAP loss) is further built to replace traditional mean squared error (mse), preserving the reconstruction clarity and structure information of HSIs, as shown in part II of Fig. 1.

In the cross-domain classification stage, with the idea of minimum class confusion (MCC) [41], a bi-classifier adversarial network (two classifiers are denoted as C_1 and C_2) is designed to optimize the confusion trend between correct

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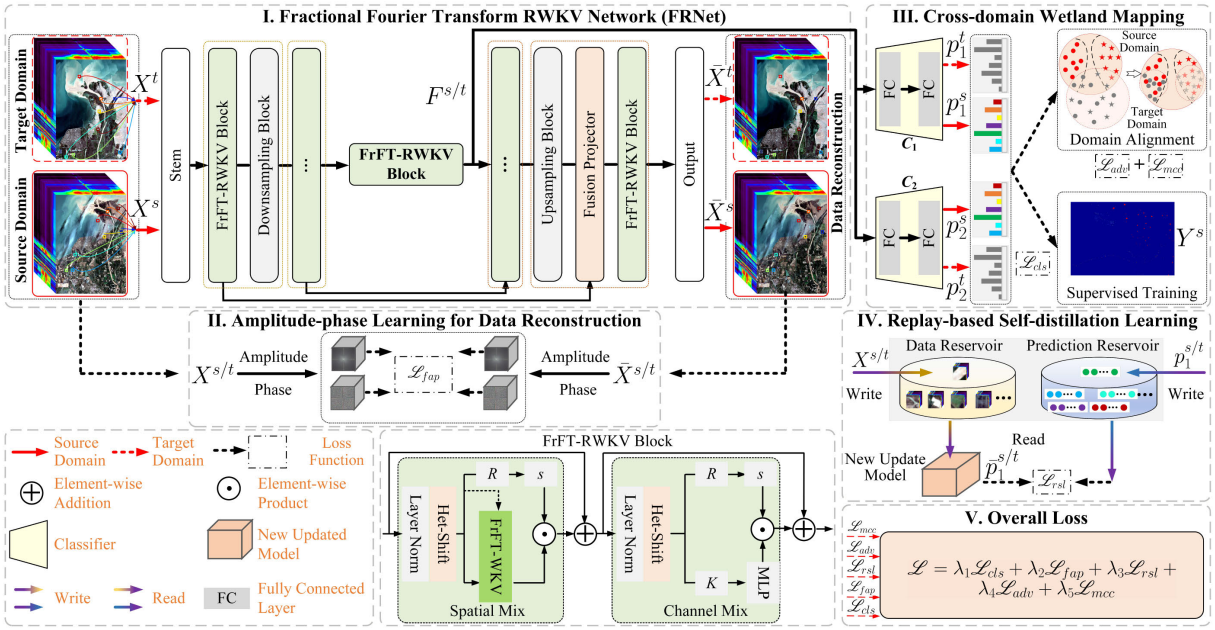


Fig. 1. Overall structure of the proposed HSI-FrFRWKV model, containing five cores: hierarchical FRNet, amplitude–phase learning for data reconstruction, cross-domain wetland mapping, replay-based self-distillation learning, and overall loss.

and ambiguous categories in the target domain to achieve implicit cross-domain alignment, as illustrated in part III of Fig. 1. Moreover, aiming at the problem of instability in adversarial training, an effective replay-based self-distillation learning strategy is devised, as shown in part IV of Fig. 1. With the above designs, the proposed HSI-FrFRWKV can generate representative domain-invariant spatial–spectral features and achieve more accurate cross-scene wetland mapping of HSIs.

FrFT-RWKV Block: The FrFT-RWKV block plays a key role in spatial–spectral–amplitude–phase feature extraction within the proposed HSI-FrFRWKV model. Different from the original RWKV block [42] that was tailored for processing 1-D sequences and failed to model global and local context in 2-D nature or multidimensional images, the designed FrFT-RWKV block addresses the issue of increased dimensionality of HSIs and better models long-term dependencies across spatial and spectral dimensions.

As shown in Fig. 1, following the structure of the original RWKV block [42], our FrFT-RWKV block is composed of a spatial mix module and a channel mix module for spatial-wise token interaction and channel-wise feature fusion, respectively. Differently, a new heterogeneous shift (Het-Shift) mechanism is designed for multiscale local information learning with a larger receptive field, while the FrFT WKV attention mechanism is further built to explore the amplitude–phase-modulated spatial–spectral information for global dependency modeling. Suppose that the input spatial–spectral features are flattened to a sequence $\bar{X} \in \mathbb{R}^{T\times c}$, where $T = h \times w$ denotes the token number and c denotes the channel dimension.

1) Spatial Mix: The spatial mix module contains two core components. The Het-Shift mechanism fuses the individual token with the shifted neighboring tokens to capture the local information, and the FrFT-WKV attention mechanism is applied to model the global dependencies of input data [42].

First, as shown in Fig. 2, the commonly used token shift mechanisms include Quad-Shift [45], Omni-Shift [46], and MVC-Shift [47]. Quad-Shift was designed with linear interpolation to fuse the current token with four adjacent tokens [45]. Omni-Shift used multibranch depth-wise convolution and structural reparameterization to fuse the current token with tokens from all directions [46]. MVC-Shift improved Omni-Shift by replacing depth-wise convolution with dilated convolution to explore the multiscale information [47]. However, these methods overlook the information in the channel (spectral) dimension during the context information learning process.

Different from the above methods, to adapt the FrFT-RWKV block to HSIs, inspired by the idea of heterogeneous convolution [43], [44], a new Het-Shift is designed. It improves the utilization of multiscale spatial information through channel (spectral) compensation under a large receptive field. Specifically, $\bar{X}$ is fed into four group convolutions with different dilation rates, whose outputs are evenly divided into four groups along the channel dimension. Then, these groups are mixed to obtain four new features, each of which is the sum of the grouped outputs with different dilation rates. These features are cascaded to obtain the shifted features $\bar{X}_{local}$

$$\tilde{X}^i = \text{split}\left(W^i_{3\times 3} *_{d^i} LN\left(\bar{X}\right)\right), \quad i = 1,2,3,4$$
$$Z = \begin{bmatrix} \tilde{X}^1_1 + \tilde{X}^2_2 + \tilde{X}^3_3 + \tilde{X}^4_4 \\ \tilde{X}^1_2 + \tilde{X}^2_3 + \tilde{X}^3_4 + \tilde{X}^4_1 \\ \tilde{X}^1_3 + \tilde{X}^2_4 + \tilde{X}^3_1 + \tilde{X}^4_2 \\ \tilde{X}^1_4 + \tilde{X}^2_1 + \tilde{X}^3_2 + \tilde{X}^4_3 \end{bmatrix} = \begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \\ Z_4 \end{bmatrix}$$
$$\bar{X}_{local} = concat\left(Z_1, Z_2, Z_3, Z_4\right) \tag{7}$$

where $\tilde{X}^i_j$ denotes the jth channel group of the output from the ith convolution, $W^i_{3\times 3} *_{d^i}$ is the weight of ith 3×3 group convolutional layer with group number $g = 4$, and dilation rate

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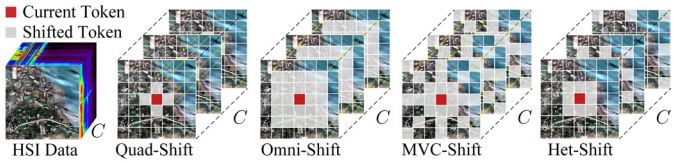


Fig. 2. Illustrations of different shift mechanisms and the proposed Het-Shift.

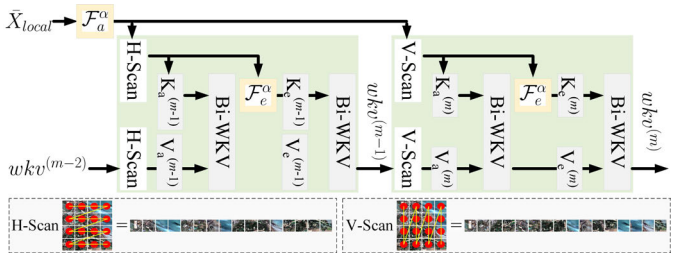


Fig. 3. Illustrations of the proposed FrFT-WKV attention mechanism.

d^i. $d^i = i$. split$(\cdot)$ denotes dividing into g groups evenly and $concat(\cdot)$ is the concatenation. $LN(\cdot)$ is the layer normalization (LN) layer. For convenience, the above process of the Het-Shift mechanism is defined as $\bar{X}_{local} = \text{Het-Shift}(LN(\bar{X}))$.

Then, the WKV attention mechanism is another core component to solve the quadratic complexity problem of self-attention in transformers [42]. In the original RWKV [42], its WKV mechanism is unidirectional and has a limited receptive field for modeling 1-D causal sequences (i.e., natural language). To enable RWKV to be suitable for noncausal data (i.e., natural images), researchers have attempted to design various improved RWKV models, such as bidirectional WKV (Bi-WKV) [45], recurrent WKV (Re-WKV) [46], [48], and 2D-WKV [47]. Conversely, HSIs contain rich spatial and high-resolution spectral information that can distinguish subtle differences between land covers in the wetlands. Directly using existing WKV schemes may restrict the improvement of classification accuracy due to the ineffective utilization of spectral information of HSIs.

As such, an FrFT-WKV attention mechanism is designed by conducting the amplitude–phase analysis and modulation of FrFT on HSIs, as shown in Fig. 3. It enables more discriminative spatial–spectral feature extraction in the optimal domain between the space and frequency domains under a global receptive field. Specifically, $\bar{X}_{local}$ is first input into a linear projection layer W_{Rs} to obtain the matrices of receptance $R_{sm} \in \mathbb{R}^{T\times c}$, and the FrFT-WKV attention mechanism by cyclically executing M times to obtain global attention map $wkv \in \mathbb{R}^{T\times c}$ from multiple directions. From Fig. 3, two different scan directions are alternated: horizontal scan (H-Scan) and vertical scan (V-Scan). The mth FrFT-WKV attention mechanism is written as

$$\begin{aligned} F_a &= \mathcal{F}_a^{\alpha}\left(\bar{X}_{local}\right) \\ K_a^{(m)} &= \Delta(F_a)W_{K_a^{(m)}}, \quad V_a^{(m)} = \Delta\left(wkv^{(m-1)}\right)W_{V_a^{(m)}} \\ K_e^{(m)} &= \mathcal{F}_e^{\alpha}(\Delta(F_a))W_{K_e^{(m)}} \\ wkv^{(m)} &= \text{Bi-WKV}\left(K_e^{(m)}, \text{Bi-WKV}\left(K_a^{(m)}, V_a^{(m)}\right)\right) \end{aligned} \tag{8}$$

where Δ is the changing direction operation and $\mathcal{F}_a^{\alpha}$ and $\mathcal{F}_e^{\alpha}$ are 1-D FrFT along the sequence and hidden dimensions, respectively. $W_{K_a^{(m)}}$, $W_{V_a^{(m)}}$, and $W_{K_e^{(m)}}$ are the linear projection layers. This design supplements the current attention result with the previous one from different scan directions for more accurate spatial–spectral–frequency representation by modeling global dependency. The above process is defined as $wkv^{(m)} = \text{FrFT-WKV}^m(\bar{X}_{local})$, $m = 1, 2, \ldots, M$. Finally, the receptance is fed into the sigmoid function $\sigma(\cdot)$ to modulate the received probability of wkv, generating output projection O_{sm} after a linear projection layer W_{Os} for the spatial mix module

$$\begin{aligned} R_{sm} &= \bar{X}_{local}W_{Rs} \\ wkv &= \text{FrFT-WKV}^{(M)}\left(\bar{X}_{local}\right) \\ O_{sm} &= \left(\sigma\left(R_{sm}\right) \odot wkv\right)W_{Os} + \bar{X}_{local} \end{aligned} \tag{9}$$

where $\odot$ means the element-wise multiplication operation.

2) Channel Mix: The channel mix module is further used for feature fusion. O_{sm} is first passed through some LN and Het-Shift layers, generating the matrices of receptance $R_{cm} \in \mathbb{R}^{T\times c}$, key $K_{cm} \in \mathbb{R}^{T\times c}$, and value $V_{cm} \in \mathbb{R}^{T\times c}$. Then, the output O_{cm} can be obtained by modulating V_{cm} using R_{cm}

$$\begin{aligned} R_{cm} &= \text{Het-Shift}(LN(O_{sm}))W_{Rc} \\ K_{cm} &= \text{Het-Shift}(LN(O_{sm}))W_{Kc} \\ V_{cm} &= \text{ReLU}(K_{cm})W_{Vc} \\ O_{cm} &= \left(\sigma\left(R_{cm}\right) \odot V_{cm}\right)W_{Oc} \end{aligned} \tag{10}$$

where W_{Rc}, W_{Kc}, W_{Vc}, and W_{Oc} are the linear projection layers and $ReLU(\cdot)$ denotes the ReLU activation function.

Unlike the existing token shift mechanisms, the devised Het-Shift mechanism not only expands the receptive field to efficiently model multiscale context, but also improves the utilization of spatial information through channel (spectral information) compensation. Moreover, the devised FrFT-WKV attention mechanism introduces frequency-domain learning methods to better mine the structural information (i.e., edges, textures, and shapes). With these two mechanisms, the newly designed FrFT-RWKV block can better explore the inherent structural characteristics under a larger receptive field, enabling representative amplitude–phase-modulated spatial–spectral feature extraction from HSIs for wetland mapping.

3) Separability of the H-Scan and V-Scan: Applying FrFT sequentially along the horizontal (H-Scan) and vertical (V-Scan) dimensions of input data is equivalent to 2-D FrFT [49].

For a given 2-D image $f(x, y)$, its 2-D FrFT is defined as

$$\mathcal{F}^{\alpha_u,\alpha_v}[f](u, v) = \int\int_{\mathbb{R}^2} K_{\alpha_u,\alpha_v}(x, y, u, v)f(x, y)\,dx\;dy \tag{11}$$

which is *separable* if kernel $K_{\alpha_u,\alpha_v}(x, y, u, v)$ can be factored

$$K_{\alpha_u,\alpha_v}(x, y, u, v) = K_{\alpha_u}(x, u) \cdot K_{\alpha_v}(y, v) \tag{12}$$

where the FrFT kernel (exponential of quadratic terms) satisfies

$$e^{i(x^2+u^2)+i(y^2+v^2)} = e^{i(x^2+u^2)} \cdot e^{i(y^2+v^2)}. \tag{13}$$

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Then, the FrFT transforms can be applied sequentially.

1) *Step 1 (H-Scan):* The 1-D FrFT $\mathcal{F}_x^{\alpha_v}$ is calculated along the x-axis of $f(x, y)$

$$g(u, y) = \mathcal{F}_x^{\alpha_u}[f](u, y) = \int_{-\infty}^{\infty} K_{\alpha_u}(x, u) f(x, y)\, dx. \quad (14)$$

2) *Step 2 (V-Scan):* $\mathcal{F}_y^{\alpha_v}$ is operated along the y-axis of $g(u, y)$

$$H(u, v) = \mathcal{F}_y^{\alpha_v}[g](u, v) = \int_{-\infty}^{\infty} K_{\alpha_v}(y, v) g(u, y)\, dy. \quad (15)$$

First, (14) is substituted into (15)

$$H(u, v) = \int_{-\infty}^{\infty} K_{\alpha_v}(y, v) \left[\int_{-\infty}^{\infty} K_{\alpha_v}(x, u) f(x, y)\, dx \right] dy \quad (16)$$

and then, the above integrals can be rearranged as

$$H(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{K_{\alpha_u}(x, u) K_{\alpha_v}(y, v)}_{K_{\alpha_u, \alpha_v}} f(x, y)\, dx\ dy. \quad (17)$$

Therefore, we can obtain

$$H(u, v) \equiv \mathcal{F}^{\alpha_u, \alpha_v}[f](u, v). \quad (18)$$

B. Cross-Domain Classification and Training Strategy

1) Amplitude–Phase Learning in the Optimal Domain Between the Space and Frequency Domains: To better activate neurons, the reconstruction strategy is utilized in the designed FRNet for spatial–spectral feature extraction, as shown in parts I and II of Fig. 1. Existing methods usually use the L_2 loss, such as the mse, as a reconstruction loss [23], [24], [50]; however, the fuzzy prediction problem is believed to be caused by this pixel-wise metric [40]. A common method is to use generative models (i.e., generative adversarial networks [51] and diffusion models [52]). Differently, a parameter-free amplitude–phase learning strategy in the FrFT domain (FAP) is designed to regularize the amplitude and phase components of predictions from FRNet, encouraging it to better focus on frequency and structure information. As shown in part II of Fig. 1, the amplitudes of input and reconstructed data are first extracted, which are regularized with mse $\mathcal{L}_{mse}$ to improve the clarity and sharpness of predictions. However, amplitude information alone is not enough for accurate generation. A straightforward idea is to directly minimize their phase difference; however, this is ineffective as the obtained phase is discontinuous. To compensate for the missing information regarding the actual shape and position, an additional phase learning method is designed by calculating the correlation between the generated and input data to learn the structure information of HSIs. The FAP loss is defined as

$$\mathcal{L}_{fap} = \gamma \mathcal{L}_{mse}\left(M(X), M\left(\hat{X}\right)\right) + (1-\gamma)\left(1 - \frac{Re\left(\langle X, \hat{X}\rangle\right)}{\sum M(X)^2 \sum M\left(\hat{X}\right)^2}\right) \quad (19)$$

where $M(\cdot)$ and $Re(\cdot)$ are the magnitude spectrum and real part of data, respectively. $\langle , \rangle$ is the complex cross correlation, $\sum$ is the summation operation, and γ is a balance factor.

2) Significance of Phase in Data Reconstruction: In (19), the phase information is utilized. By maximizing the real part of the cross correlation, which requires phase alignment, the structural information of HSIs can be effectively preserved.

First, the above complex cross correlation term C can be derived from the inner product

$$C = \langle X, \hat{X}\rangle = \int_{-\infty}^{\infty} X(u)\hat{X}^*(u)\, du \quad (20)$$

which is then substituted into the polar form

$$C = \int_{-\infty}^{\infty} M_X(u) e^{i\theta_X(u)} \cdot M_{\hat{X}}(u) e^{-i\theta_{\hat{X}}(u)}\, du = \int_{-\infty}^{\infty} M_X(u) M_{\hat{X}}(u) e^{i\left(\theta_X(u) - \theta_{\hat{X}}(u)\right)}\, du \quad (21)$$

where $X(u) = M_X(u)e^{i\theta_X(u)}$ and $\hat{X}(u) = M_{\hat{X}}(u)e^{i\theta_{\hat{X}}(u)}$ are the input and reconstructed data in the frequency domain. $*$ means the complex conjugate and θ is the phase angle.

Then, with Euler's formula $e^{ix} = \cos x + i \sin x$, the following equation can be updated as:

$$C = \int_{-\infty}^{\infty} M_X(u) M_{\hat{X}}(u) \left[\cos(\Delta\theta) + i \sin(\Delta\theta)\right] du \quad (22)$$

where $\Delta\theta(u) = \theta_X(u) - \theta_{\hat{X}}(u)$. Equation (19) aims to maximize the similarity of phase, corresponding to maximizing $\mathrm{Re}(C)$

$$\mathrm{Re}(C) = \int_{-\infty}^{\infty} M_X(u) M_{\hat{X}}(u) \cos\left(\theta_X(u) - \theta_{\hat{X}}(u)\right) du. \quad (23)$$

Assuming magnitudes M_X and $M_{\hat{X}}$ are nonnegative. Equation (23) is maximized when the third term is maximized at u

$$\max \implies \cos\left(\theta_X(u) - \theta_{\hat{X}}(u)\right) = 1 \quad \forall u \quad (24)$$

thereby obtaining

$$\theta_X(u) = \theta_{\hat{X}}(u). \quad (25)$$

Therefore, in (19), minimizing the phase learning loss can force the predicted phase $\theta_{\hat{X}}$ to converge to the true phase θ_X. Since the phase encodes spatial structure (i.e., edges and shapes) of HSIs, this guarantees structural preservation.

3) Cross-Domain Alignment: Furthermore, to align the source and target domains, the bi-classifier adversarial network [53] is introduced, followed by an MCC loss [41] to reduce confusion between the two classifiers for different classes in the target domain. From part III of Fig. 1, compared with other DA methods [22], [23], [24], this method can not only determine whether the features belong to the source or target domain, but also enable the task-specific decision boundaries between different classes.

Specifically, a three-step adversarial training strategy [53] is used for cross-domain alignment. First, the feature generator (FRNet) denoted as G and two classifiers (denoted as C_1 and C_2) are jointly optimized using the classification loss $\mathcal{L}_{cls}$ on the source predictions p_1^s and p_2^s, the MCC loss $\mathcal{L}_{mcc}$ on the target predictions p_1^t and p_2^t, and the reconstruction loss $\mathcal{L}_{fap}$ on the source-target samples, enabling the whole model to learn initial representations. Second, G is fixed while C_1 and C_2 are updated to maximize their predictions p_1^t and p_2^t on the target-domain samples with the absolute value metric (denoting as $\mathcal{L}_{adv}$), helping identify ambiguous target

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samples near the decision boundary. Third, C_1 and C_2 are fixed, and G is optimized to minimize their predictions with $\mathscr{L}_{mcc}$ and $\mathscr{L}_{adv}$, encouraging target-domain features to move toward source-supported category-discriminative regions. By iteratively performing these three steps, an adversarial learning process is established between the feature generator and classifiers, leading to fine-grained category-level domain alignment.

4) Replay-Based Self-Distillation Learning: Generally, when neural networks are trained on new data, they will quickly forget what they have learned before, known as catastrophic forgetting [54], [55]. An important mechanism for protecting memory in the human brain is to reactivate the neuronal activity patterns that represent these memories [56]. As such, a replay-based self-distillation learning strategy is designed to alleviate the instability of the adversarial training.

From part V of Fig. 1, first, during the writing process, the data reservoir T_x and prediction reservoir T_c are defined to store the input data $X^{s/t}$, and their predictions $p^{s/t}$ from the classifier C_1 of the first n iterations. Then, during the reading process, $X^{s/t}$ and $p^{s/t}$ of a certain iteration are randomly read, and the new predictions $\bar{p}^{s/t}$ of $X^{s/t}$ are obtained by using the updated model. Finally, during the self-distillation process, the Softmax function with temperature coefficient τ is used to convert these two predicted results into the probability distributions to obtain the corresponding soft labels $y^{s/t} = Softmax((p^{s/t}/\tau))$ and $\bar{y}^{s/t} = Softmax((\bar{p}^{s/t}/\tau))$. The Kullback–Leibler (KL) divergence loss $\mathscr{L}_{KL}(\cdot)$ is further used as an auxiliary consistency loss to force the new prediction result to be consistent with the previous one across different batches

$$\mathscr{L}_{rsl} = \mathscr{L}_{KL}\left(y^{s/t}, \bar{y}^{s/t}\right). \tag{26}$$

C. Overall Loss

From Fig. 1, there are five losses, containing amplitude–phase reconstruction loss ($\mathscr{L}_{fap}$) in part II, supervised classification loss ($\mathscr{L}_{cls}$) and cross-domain alignment losses ($\mathscr{L}_{adv}$ and $\mathscr{L}_{mcc}$) in part III, and self-distillation loss ($\mathscr{L}_{rsl}$) in part IV. The overall loss $\mathscr{L}$ of our HSI-FrFRWKV is written as

$$\mathscr{L} = \lambda_1\mathscr{L}_{cls} + \lambda_2\mathscr{L}_{fap} + \lambda_3\mathscr{L}_{rsl} + \lambda_4\mathscr{L}_{adv} + \lambda_5\mathscr{L}_{mcc} \tag{27}$$

where λ_j is the balance factors, and $j = 1, 2, 3, 4, 5$, $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5 = 1$ in all experiments.

The overall loss $\mathscr{L}$ is optimized through the above three-step alternating adversarial training strategy, where the auxiliary consistency loss $\mathscr{L}_{rsl}$ will be integrated into the first step. With the designed FRNet and the above training strategies, the proposed HSI-FrFRWKV model can not only achieve the desired reconstruction clarity and explore the actual shape and position of HSIs, but also relieve problems of catastrophic forgetting and unstable training. Therefore, more accurate and domain-invariant spatial–spectral–frequency features can be extracted from HSIs for cross-scene wetland mapping.

IV. Experimental Results

One public cross-domain HSI dataset including Houston 2013 to Houston 2018 (H13–H18), and two cross-domain wetland HSI datasets including Ziyuan (ZY)1-02D Yancheng to Gaofen (GF)-5 Yangcheng (YC–YC), and ZY1-02D Huanghekou to ZY1-02D Huanghekou (HHK–HHK) are used. Overall accuracy (OA), average accuracy (AA), and Kappa (κ) coefficient are used as the quantitative metrics, whose average values after five runs are reported. All experiments are conducted using an Nvidia GeForce RTX 3080 GPU and Intel Core i7-12700F processor with PyTorch 2.0 and Python 3.10.

A. Experimental Data

1) H13–H18 Dataset: It is composed of Houston 2013 (source) and Houston 2018 (target) data, which were collected by different sensors over the University of Houston, Texas, USA. Source data contain 349 × 1905 pixels at a spatial resolution of 2.5 m and 144 bands, while target data contain 210 × 954 pixels in a finer spatial resolution of 1 m and 48 bands. 7 consistent classes and 48 overlapped bands are selected for cross-scene analysis. Fig. 4(a)–(c) illustrates its false-color maps, ground-truth maps, and sample number.

2) YC–YC Dataset: As shown in [1], this dataset contains ZY1-02D Yancheng (source) data and GF-5 Yancheng (target) data, which were captured by the AHSI aboard China's ZY1-02D and GF-5 satellites, respectively. Their image sizes are 1398 × 942 and 1175 × 585 pixels, respectively, and there are 147 spectral bands and 7 consistent land covers selected for cross-sensor analysis. Their false-color maps, ground-truth maps, and number of samples are shown in Fig. 4(d)–(f).

3) HHK–HHK Dataset: According to [1], with the ZY1-02D-AHSI, two ZY1-02D Huanghekou datasets were collected on June 28, 2020, and September 29, 2021, respectively. The former is the source domain with an image size of 1147 × 1600 pixels, while the latter is the target domain with 1050 × 1219 pixels. There are 108 spectral bands and 8 land covers for cross-domain analysis. Its false-color maps, ground-truth maps, and sample information are shown in Fig. 4(g)–(i).

B. Experimental Settings

This section designs four experiments to verify the performance of the proposed HSI-FrFRWKV model.

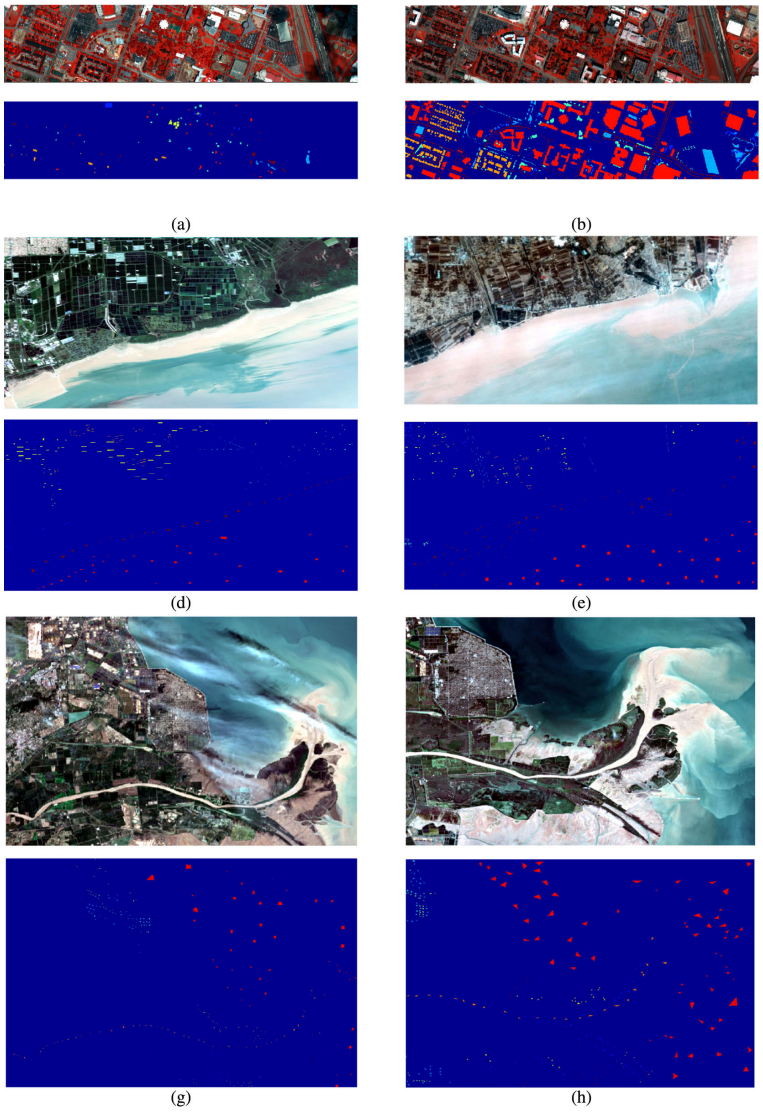
1) Parameter Setting Analysis: The first experiment is designed for parameter analysis, including fractional order α, number M of changing direction, and learning rate l_r, whose candidate sets are defined as $\alpha \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$, $M \in \{1, 2, 3, 4\}$, $l_r \in \{0.01, 0.05, 0.1, 0.5\}$, respectively.

2) Classification Performance Analysis: The second experiment is conducted to compare HSI-FrFRWKV with six competitive models, containing SSWADA [1], FDDAN [15], FDGNet [31], S^4DL [20], ADNet [21], and DHSNet [27].

3) Ablation Study: The third experiment is built to validate the contributions of some new blocks (i.e., FrFT-RWKV block, FAP loss, Het-Shift mechanism, and replay-based self-distillation strategy) on cross-scene wetland mapping performance.

4) Complexity Analysis: The four experiments are devised for a detailed complexity analysis of the proposed HSI-FrFRWKV framework on three cross-scene HSI datasets.

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Houston 13 to Houston 18

Class			Number of Samples	
NO.	Color	Name	Source	Target
1		Grass healthy	345	1353
2		Grass stressed	365	4888
3		Trees	365	2766
4		Water	285	22
5		Residential buildings	319	5347
6		Non-commercial buildings	408	32459
7		Roads	443	6365
Total			2530	53200

(c)

ZY1-02D Yancheng to GF-5 Yancheng

Class			Number of Samples	
NO.	Color	Name	Source	Target
1		Architecture	451	360
2		River	214	217
3		Reed	244	132
4		Paddy	3026	832
5		Fallow Land	650	234
6		Sea	2076	2395
7		Offshore Water	1558	1305
Total			8219	5475

(f)

ZY1-02D Huanghekou (2020) to Huanghekou (2021)

Class			Number of Samples	
NO.	Color	Name	Source	Target
1		Reed	310	1057
2		Salt Flat Filtration Pond	247	512
3		Salt Flat Evaporation Pond	300	260
4		Salt Falt	306	412
5		Suaeda	218	418
6		River	584	941
7		Sea	4694	10239
8		Tide Ditch	67	248
Total			6726	14087

(i)

Fig. 4. (Top) False-color map of HSIs and (bottom) ground-truth map. (a)–(c) H13–H18 dataset (HSI with Rband: 35, Gband: 15, Bband: 10). (d)–(f) YC–YC dataset (target HSI with Rband: 20, Gband: 10, Bband: 5). (g)–(i) HHK–HHK dataset (HSI with Rband: 20, Gband: 10, Bband: 5).

C. Parameter Setting Analysis

Fig. 5 reports OA values to analyze the effects of some parameters on cross-scene wetland mapping of HSI-FrFRWKV.

1) Analysis of Fractional Order α: α denotes the fractional order, whose effects on cross-domain performance are shown in Fig. 5(a). The OA values on the three datasets fluctuate up and down with the value of α. Particularly, we also conduct experiments to analyze the way α is valued by comparing whether it is fixed or learnable, where using a fixed value based on specific datasets has some performance advantages.

2) Analysis of Number of Changing Direction M: M is the number of changing directions. From Fig. 5(b), based on the complexity and OA value, M is set to 1 for the three datasets.

3) Analysis of Learning Rate l_r: l_r determines the degree of parameter adjustments. Its value being too large or too small can lead to slow convergence or poor generalization. From Fig. 5(c), the best performance occurs when $l_r = 0.1$.

D. Classification Performance Analysis

The classification results (i.e., class-specific accuracy, OA, AA, and κ) of the proposed HSI-FrFRWKV and other six comparison methods for the three datasets are reported in Tables I–III, with their classification maps shown in Figs. 6–8.

1) Structure Analysis: From the perspective of distribution alignment, FDDAN designed two disentanglers to process the global–local mix features, followed by adversarial learning for cross-domain classification; S^4DL utilized a gradient-guided spatial–spectral decomposition, followed by a shift-sensitive adaptive monitor to modify the alignment strategy for various domain shifts; FDGNet introduced the data geometry into extension domain, with a sampling regime for domain-invariant representation; ADNet devised a homogeneous dual-branch encoder, followed by domain-specific instance shuffling and multiattribute representation learning for semantic gap reduction. From the perspective of boundary adjustment of classifiers, SSWADA introduced weighted

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TABLE I
CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS FOR THE H13–H18 DATASET

Class	SSWADA	FDDAN	FDGNet	S^{4}DL	ADNet	DHSNet	Proposed
1	31.38 ± 23.77	80.62 ± 18.37	79.75 ± 8.35	78.37 ± 16.26	59.29 ± 6.09	60.41 ± 19.34	**85.63 ± 11.35**
2	83.34 ± 8.29	61.64 ± 18.99	69.69 ± 4.86	82.35 ± 11.12	**86.66 ± 6.22**	79.89 ± 9.01	70.47 ± 19.49
3	51.29 ± 9.54	62.97 ± 8.23	48.73 ± 6.75	54.06 ± 4.78	65.13 ± 4.50	63.82 ± 4.53	**65.96 ± 3.31**
4	96.36 ± 5.93	93.64 ± 6.10	**100.00 ± 0.00**	98.18 ± 4.07	**100.00 ± 0.00**	**100.00 ± 0.00**	93.64 ± 6.10
5	68.39 ± 16.30	64.66 ± 8.31	69.15 ± 3.08	66.61 ± 11.96	64.66 ± 8.31	75.28 ± 17.26	**81.20 ± 9.39**
6	82.79 ± 6.90	81.05 ± 4.21	**88.02 ± 1.23**	73.69 ± 2.11	85.69 ± 3.45	87.70 ± 4.47	87.43 ± 3.50
7	49.61 ± 20.21	59.06 ± 15.63	55.18 ± 2.31	64.48 ± 3.54	59.49 ± 9.14	64.21 ± 5.43	**74.66 ± 7.20**
OA	74.48 ± 0.97	75.48 ± 0.62	78.26 ± 0.29	71.78 ± 1.24	78.80 ± 0.48	80.99 ± 1.46	**82.56 ± 1.02**
AA	66.17 ± 7.34	74.00 ± 1.83	72.93 ± 1.64	73.96 ± 3.19	74.41 ± 2.50	75.90 ± 6.05	**79.86 ± 2.43**
$\kappa \times 100$	57.99 ± 4.11	60.21 ± 1.45	62.87 ± 0.52	57.15 ± 1.89	64.61 ± 0.89	68.00 ± 3.83	**71.51 ± 1.63**

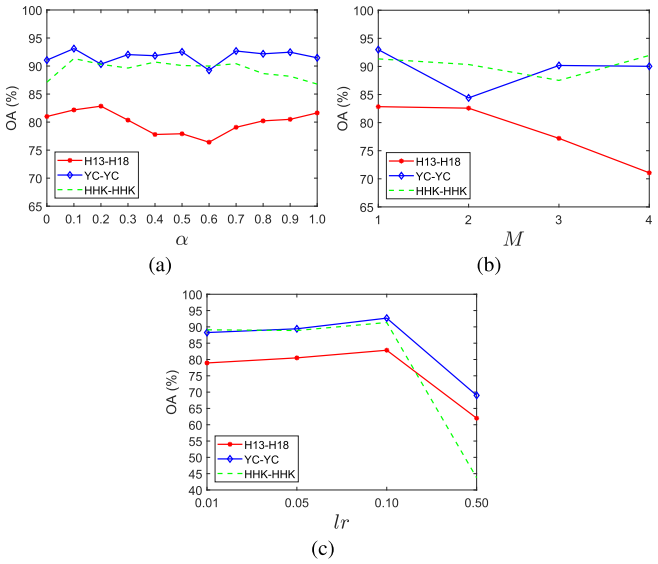


Fig. 5. Parameter analysis for HSI-FrFRWKV under three datasets. (a) Parameter analysis for fractional order α. (b) Parameter analysis for changing direction M. (c) Parameter analysis for learning rate l_r.

adversarial discrimination for source instance weighting and a multiclassifier for target-specific information learning; DHSNet built a dual classifier and a self-training strategy to reduce domain gaps while preventing error accumulation of pseudo-labels.

Differently, from the perspective of amplitude–phase analysis, our HSI-FrFRWKV integrated FrFT into RWKV with linear complexity for spatial–spectral–frequency feature extraction. Moreover, the FAP loss enables the above process in the optimal domain between the space and frequency domains for clarity and structure preservation of reconstructed data. A replay-based self-distillation learning method is further designed to relieve unstable training and catastrophic forgetting, thus achieving better cross-domain wetland mapping on HSIs.

2) Results and Analysis on the H13–H18 Dataset: Table I reports the classification results of HSI-FrFRWKV and six comparison methods in the H13–H18 dataset. Among all comparison methods, their OA value ranges from 71.78% to 80.99%, with the highest AA value being only 75.90%. Specifically, HSI-FrFRWKV achieves the best cross-domain performance, with 1.57% gains in the OA value, 3.96% in the AA value, and 3.51% gains in the κ value compared to the second-best method. In addition, its classification accuracy in each category exceeded 65.96%. Moreover, based on the visualization results in Fig. 6, our method achieves more refined classification results, which are closer to the ground-truth map. Experimental results verify the effectiveness of the proposed HSI-FrFRWKV.

3) Results and Analysis on the YC–YC Dataset: Table II compares HSI-FrFRWKV with respect to other competitors on the YC–YC dataset. It is evident that the OA values of all methods can reach above 85.40%. Among them, they achieve the best performance in class 6 ("sea"), where the classification accuracy of FDGNet, S^{4}DL, ADNet, AHSNet, and HSI-FrFRWKV all reach 100%, while SSWADA and FDDAN reach 99.99% and 97.95%, respectively. As for class 2 ("river"), the classification accuracy of all six comparison methods is below 63.41%, while HSI-FrFRWKV has a much better classification accuracy than them, reaching up to 92.90%. Particularly, due to the small number of samples in class 2 and class 5, most methods have low classification accuracy, with SSWADA and FDDA even having only 1.21% and 0%, respectively. It is precisely because of the designed FrFT-RWKV block and replay-based self-distillation learning strategy that HSI-FrFRWKV achieves optimal results in three metrics, achieving gains of 1.87%, 3.07%, and 2.54% compared to their optimal values, respectively. Based on the classification maps in Fig. 7, the visualization effect of HSI-FrFRWKV is far superior to other methods, with the best category boundary and overall classification performance. Experimental results also verify the superiority of HSI-FrFRWKV for cross-domain wetland mapping on HSIs.

4) Results and Analysis on the HHK–HHK Dataset: The classification results and maps of all methods on the HHK–HHK dataset are shown in Table III and Fig. 8, respectively. Compared to the other two datasets, this dataset is more difficult to classify due to uneven sample distribution and fragmented distribution of land covers, resulting in low AA values for most methods. The OA value of all comparison methods is below 89.00%, with only HSI-FrFRWKV exceeding 90.00%. In addition, FDGNet and ADNet are unable to accurately classify classes 2, 4, 5, and 8, with an accuracy of less than 17.96%, which is the main reason why their AA values are, respectively, only 41.82% and 42.07%. Particularly, HSI-FrFRWKV illustrates the best performance in OA, AA, and κ, reaching 90.02%, 70.05%, and 78.34%, respectively, which is consistent with the results in Fig. 8. Experimental results further verify the excellent cross-domain wetland mapping performance of the proposed HSI-FrFRWKV model.

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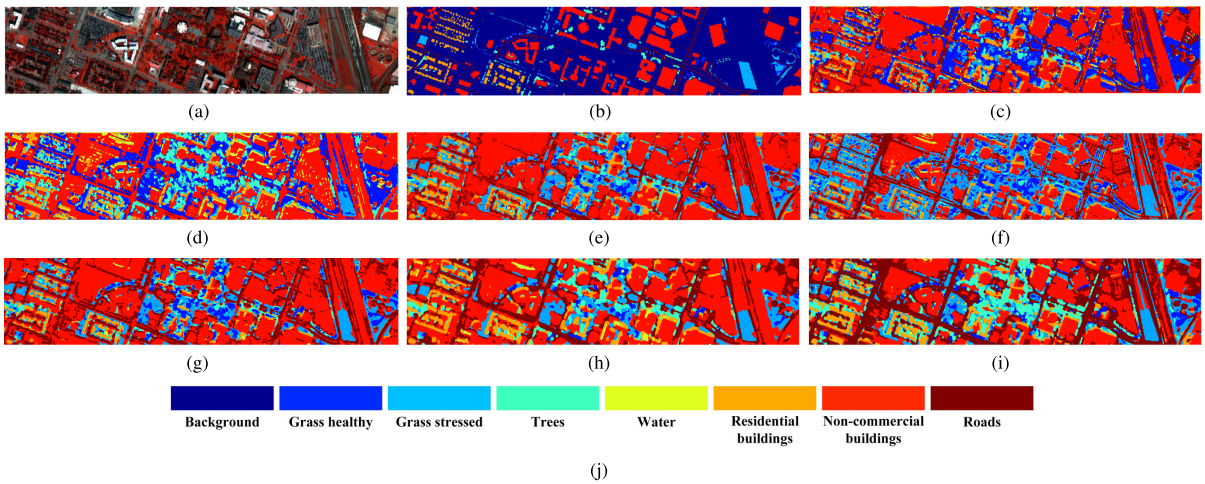


Fig. 6. Classification maps for the H13–H18 dataset. (a) Target HSI (Rband: 35, Gband: 15, Bband: 10). (b) Target ground-truth map. (c) SSWADA. (d) FDDAN. (e) FDGNet. (f) S^{4}DL. (g) ADNet. (h) DHSNet. (i) Proposed. (j) Legend.

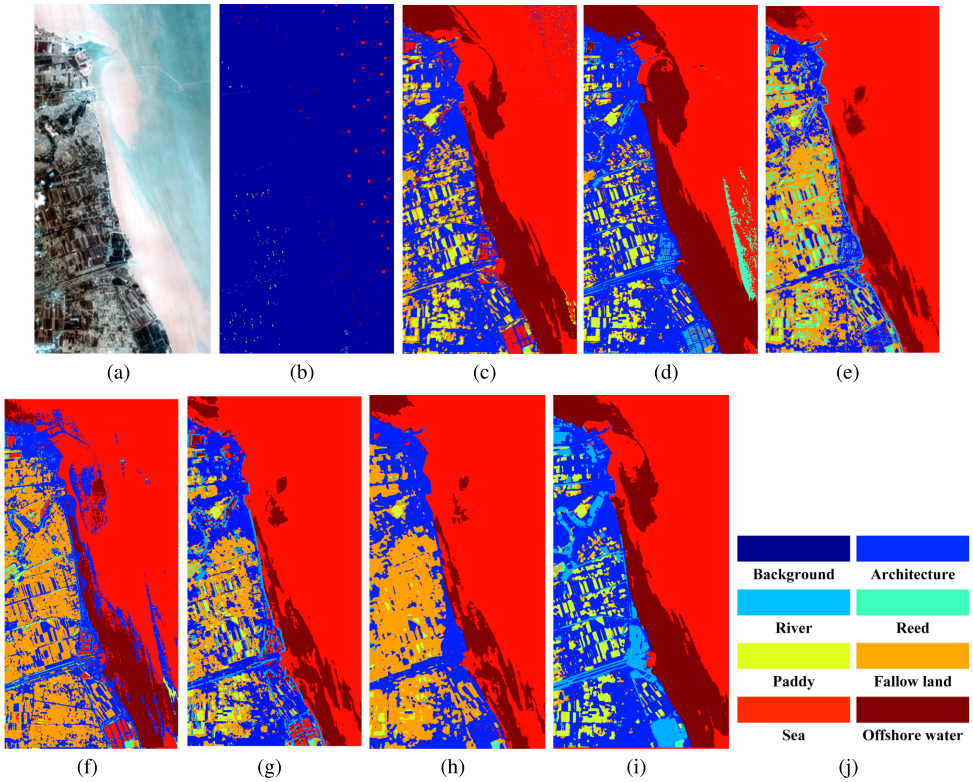


Fig. 7. Classification maps for the YC–YC dataset. (a) Target HSI (Rband: 20, Gband: 10, Bband: 5). (b) Target ground-truth map. (c) SSWADA. (d) FDDAN. (e) FDGNet. (f) S^{4}DL. (g) ADNet. (h) DHSNet. (i) Proposed. (j) Legend.

E. Ablation Study

Some key components (i.e., FrFR-RWKV block, Het-Shift mechanism, FAP loss, and replay-based self-distillation learning) are designed to improve spatial–spectral feature extraction and cross-domain wetland mapping performance of HSI-FrFRWKV. A detailed ablation study is designed to assess their roles using OA and κ values, as shown in Table IV. The proposed model with and without each component are written as the proposed (w/) and the proposed (w/o), respectively.

1) Analysis of the Het-Shift Mechanism: Part (a) of Table IV reports the OA and κ values of existing token shift methods (including Quad-Shift, Omni-Shift, and MVC-Shift) and our Het-Shift mechanism. Quad-Shift fuses the four adjacent tokens through a linear method, which fails to learn multiscale spatial information, resulting in poor performance, with OA values of only 77.41%, 90.30%, and 86.92%, respectively. Omni-Shift adopts multidirectional extension to capture contextual information in all directions, increasing OA values by 3.33%, 0.69%, and 2.08%, respectively. MVC-Shift introduces dilated convolution to explore multiscale spatial information with a larger receptive field, achieving at least 0.49% gains in OA value compared to the above methods.

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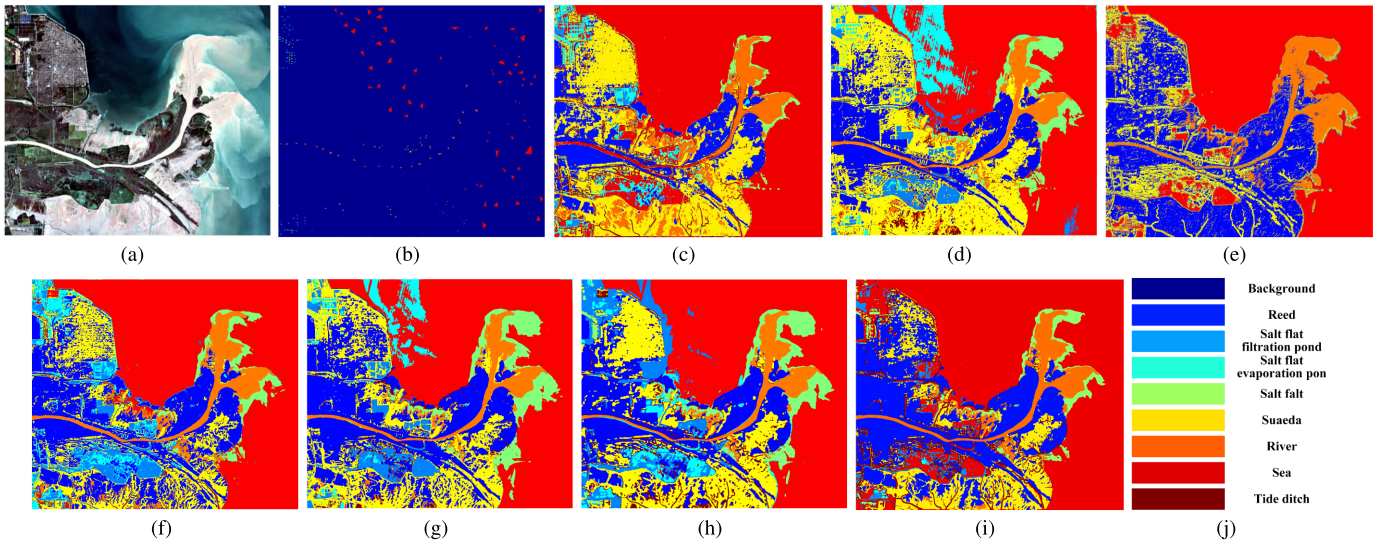


Fig. 8. Classification maps for the HHK–HHK dataset. (a) Target HSI (Rband: 20, Gband: 10, Bband: 5). (b) Target ground-truth map. (c) SSWADA. (d) FDDAN. (e) FDGNet. (f) S^4DL. (g) ADNet. (h) DHSNet. (i) Proposed. (j) Legend.

TABLE II
CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS FOR THE YC–YC DATASET

Class	SSWADA	FDDAN	FDGNet	S^4DL	ADNet	DHSNet	Proposed
1	85.17 ± 15.56	88.11 ± 15.72	96.39 ± 3.00	75.89 ± 15.65	89.56 ± 9.76	94.17 ± 5.13	**96.67 ± 4.70**
2	35.94 ± 26.56	63.41 ± 34.76	15.30 ± 10.01	15.48 ± 18.30	25.35 ± 15.75	14.29 ± 14.30	**92.44 ± 11.86**
3	1.21 ± 2.71	00.00 ± 0.00	**54.55 ± 20.82**	13.48 ± 17.73	25.00 ± 23.16	50.30 ± 23.25	28.18 ± 25.07
4	99.01 ± 0.61	92.02 ± 4.71	84.04 ± 6.85	93.13 ± 6.61	88.92 ± 6.30	80.07 ± 16.43	**99.42 ± 0.42**
5	29.83 ± 33.98	52.65 ± 39.06	36.58 ± 21.83	**86.07 ± 15.64**	41.28 ± 21.41	80.85 ± 18.56	12.99 ± 22.29
6	99.99 ± 0.02	97.95 ± 2.80	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
7	**99.33 ± 0.44**	97.67 ± 3.13	82.65 ± 7.95	98.51 ± 3.07	79.23 ± 7.71	83.97 ± 7.55	95.42 ± 2.57
OA	90.79 ± 1.05	90.67 ± 2.01	86.04 ± 1.90	90.98 ± 1.17	85.40 ± 0.82	87.35 ± 4.31	**92.85 ± 0.53**
AA	64.35 ± 4.35	70.26 ± 5.56	67.07 ± 2.90	68.94 ± 4.56	64.19 ± 5.08	71.95 ± 6.56	**75.02 ± 2.59**
$\kappa \times 100$	87.13 ± 1.48	87.10 ± 2.75	80.42 ± 2.73	87.46 ± 1,61	79.42 ± 1.31	82.30 ± 6.04	**90.00 ± 0.75**

The designed Het-Shift mechanism can effectively explore the multiscale context information and enhance the utilization of spatial information through channel (spectral) compensation under a large receptive field. Compared with the above three methods, it increases OA and κ values by at least 0.53% and 1.58% on the three datasets, respectively, illustrating its effectiveness.

2) Analysis of the FrFT-RWKV Block and the FAP Loss: The FrFT-RWKV block and FAP loss play a key role in the extraction of spatial–spectral–frequency features. We design some experiments in part (b) of Table IV to compare the performance with and without them. From the first line of part (b), by removing the FAP loss and FrFT-RWKV block, the OA values are reduced from 82.56% to 78.45% and 92.85% to 89.99% and 90.02% to 87.26%. From the second line of part (b), by introducing the FrFT-RWKV block with mse loss, the OA value increases by at least 1.67%. Furthermore, by analyzing part (d) and the second-to-last row of part (b), the average OA value of the proposed model decreases by 0.99% using only FAP (w/o amplitude) loss, while using only FAP (w/o phase) loss results in a decrease of at least 0.95%. By combining amplitude and phase analysis, the proposed (w/ FAP) achieves an average gain of 0.77% and 1.74% in OA and κ compared to the proposed (w/ mse), respectively. These results prove the advantages of the FrFT-RWKV block and the FAP loss.

3) Analysis of the Replay-Based Self-Distillation Strategy: To alleviate the problems of model instability and catastrophic forgetting during the training process, a replay-based self-distillation learning strategy is designed. From the experimental results in part (c) of Table IV, after removing it, the OA value of the proposed model decreases by 1.88%, 0.92%, and 1.23% on three datasets, respectively, verifying its importance for model training and cross-scene classification performance.

4) Qualitative Analysis of Components: To intuitively analyze these new components, we further provide gradient-weighted class activation mapping (Grad-CAM). In particular, proposed (w/o FrFT-RWKV) means that FrFT-RWKV is replaced with vanilla RWKV, proposed (w/o FAP) denotes that FAP is replaced with mse, and the proposed (w/o all) represents that all new components are removed, using RWKV and mse. From Fig. 9, the Grad-CAM response of our HSI-FrFRWKV (red area) is highly concentrated at the center of the input patch for land cover, and aligned with its boundary (yellow outline). In contrast, after removing the newly designed components one by one, the responses at the same location are clearly dispersed into the background or another category area, where the cross-scene performance is the worst after removing all components, as shown in Fig. 9(d) and (f). These

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TABLE III

CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS FOR THE HHK–HHK DATASET

Class	SSWADA	FDDAN	FDGNet	S^{4}DL	ADNet	DHSNet	Proposed
1	85.54 ± 10.47	94.95 ± 5.74	78.30 ± 32.28	93.04 ± 6.25	**99.47 ± 0.88**	82.42 ± 25.74	89.71 ± 15.46
2	22.93 ± 9.28	**34.57 ± 4.79**	4.30 ± 7.14	32.85 ± 7.58	5.51 ± 11.26	36.64 ± 21.56	15.55 ± 1.83
3	**92.23 ± 6.83**	44.07 ± 16.32	24.61 ± 36.67	66.00± 16.48	6.08 ± 6.10	19.62 ± 33.40	28.38 ± 31.95
4	97.96 ± 3.33	85.92 ± 16.00	17.96 ± 19.06	71.60 ± 10.80	7.57 ± 14.69	**99.85 ± 0.33**	91.84 ± 8.47
5	35.55 ± 8.19	32.15 ± 22.84	10.57 ± 20.59	24.83 ± 10.38	17.70 ± 20.99	37.32 ± 3.86	**52.01 ± 21.30**
6	66.80 ± 44.20	99.40 ± 0.80	96.79 ± 2.34	99.30± 1.51	93.26 ± 13.66	**99.89 ± 0.24**	99.57 ± 0.95
7	**96.32 ± 1.71**	86.70 ± 4.34	95.16 ± 3.29	95.75 ± 1.23	95.05 ± 0.84	88.33 ± 4.45	96.00 ± 1.01
8	47.02 ± 34.36	60.40 ± 35.99	6.86 ± 12.23	29.44 ± 25.07	11.93 ± 26.46	77.66 ± 8.98	**87.34 ± 2.97**
OA	88.17 ± 1.45	83.38 ± 2.74	83.07 ± 2.83	88.97 ± 1.28	84.05 ± 0.47	84.15 ± 3.44	**90.02 ± 1.17**
AA	68.04 ± 2.75	67.27 ± 7.87	41.82 ± 5.22	64.10 ± 4.38	42.07 ± 2.76	67.72 ± 4.30	**70.05 ± 2.36**
κ	73.43 ± 4.48	68.37 ± 4.16	62.64 ± 3.64	76.48 ± 2.54	65.35 ± 0.95	68.86 ± 6.01	**78.34 ± 2.36**

TABLE IV

ABLATION STUDY OF THE PROPOSED HSI-FRFRWKV MODEL ON THE THREE CROSS-DOMAIN HSI DATASETS

	Components				H13-H18 Dataset		YC-YC Dataset		HHK-HHK Dataset	
	Quad-Shift	Omni-Shift	MVC-Shift	Het-Shift	OA	$\kappa \times 100$	OA	$\kappa \times 100$	OA	$\kappa \times 100$
(a)	✔	✘	✘	✘	77.41	63.51	90.30	86.43	86.92	71.84
	✘	✔	✘	✘	80.74	68.01	90.99	87.43	89.00	76.67
	✘	✘	✔	✘	81.66	69.45	91.54	88.15	89.49	76.76
	FrFT-RWKV	FAP(w/o amplitude)	FAP(w/o phase)							
(b)	✘	✘	✘	✔	78.45	63.45	89.99	85.92	87.26	71.50
	✔	✘	✘	✔	81.94	69.72	92.15	88.98	88.93	75.76
	✔	✔	✘	✔	81.89	70.47	91.34	87.88	89.11	75.96
	✔	✘	✔	✔	80.87	69.04	91.71	88.37	89.07	75.61
(c)	Proposed (w/o replay)				80.68	68.40	91.93	88.69	88.79	75.35
(d)	Proposed				**82.56**	**71.51**	**92.85**	**90.00**	**90.02**	**78.34**

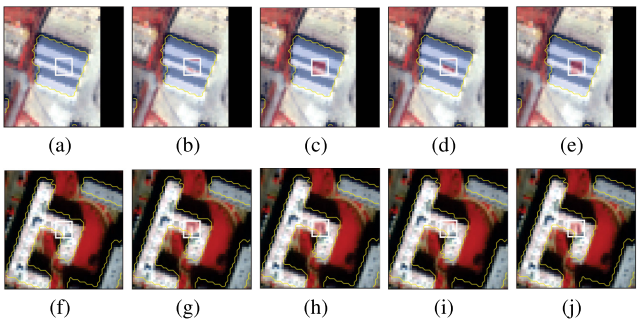


Fig. 9. Grad-CAM maps for the H13–H18 dataset. (a) and (f) False-color map for some input patches. (b) and (g) Proposed (w/o FrFT-RWKV). (c) and (h) Proposed (w/o FAP). (d) and (i) Proposed (w/o all). (e) and (j) Proposed.

results indicate that these components can force the whole model to focus on discriminative regions with high semantic consistency, further qualitatively verifying their effectiveness and superiority.

F. Complexity Analysis

The last experiment is built for a detailed complexity analysis of the proposed HSI-FrFRWKV framework.

1) Complexity Analysis Between Different Methods: The complexity of different methods is first analyzed, with the OA, κ, training time per epoch $(Tr_time)(s)$, testing time $(Te_time)(s)$, parameters (K), and FLOPs (M) reported in Table V. Compared with the SSWADA, FDDAN, FDGNet, S^{4}DL, ADNet, and DHSNet, the classification performance and parameters of our HSI-FrFRWKV on the three datasets have absolute advantages and have some advantages over several comparison methods in terms of training time, testing time, and FLOPs. After removing FrFT, the proposed (w/o FrFT) shows significant superiority in testing time and FLOPs, outperforming most comparison methods, but inevitably loses some classification accuracy (still competitive compared to other comparison methods). The experimental results demonstrate the advantages of the proposed HSI-FrFRWKV in terms of classification performance and complexity under two working modes (whether FrFT is used or not).

2) Complexity Analysis Between Different Components: We further compare the accuracy and complexity of RWKV and convolutional neural network (CNN)/Transformer-based methods. From Table V, although CNN leads the proposed (w/ Transformer), the proposed (w/o FrFT) (i.e., original RWKV), and the proposed (w/ FrFT) in complexity, there is a significant gap in accuracy metrics between them, especially in complex scenarios. This is also the main reason why the majority of current methods do not use CNN in their backbone design. Moreover, compared with the proposed (w/ Transformer), the proposed (w/o FrFT) has advantages in almost all metrics, except for some disadvantages in FLOPs. After introducing FrFT, the entire model only has absolute advantages in accuracy and parameters, which is also an inevitable result of considering theoretical design and classification performance comprehensively in the proposed HSI-FrFRWKV model.

In summary, compared to other comparison methods and commonly used CNNs/Transformers, the proposed HSI-FrFRWKV achieves a good balance between classification performance and complexity on the task of cross-scene HSI wetland mapping, verifying its effectiveness and superiority.

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TABLE V
COMPLEXITY ANALYSIS OF DIFFERENT METHODS ON THE THREE DATASETS

Methods	H13-H18 Dataset					
	OA (%)	κ (%)	Tr_Time (s)	Te_Time (s)	Parameters (K)	Flops (M)
SSWADA	74.48	57.99	2.20	19.64	14315.72	14.32
FDDAN	75.48	60.21	12.10	37.77	2395.28	20.22
FDGNet	78.26	62.87	1.27	34.60	1815.38	139.50
S^4DL	71.78	57.15	1.10	19.75	1561.56	14.15
ADNet	78.80	64.61	1.32	9.77	1481.22	106.55
DHSNet	80.99	68.00	0.37	8.62	6509.87	18.47
Proposed (w/ CNN)	75.59	59.95	3.47	16.45	541.92	18.26
Proposed (w/ Transformer)	77.46	62.91	4.07	21.22	941.41	10.85
Proposed (w/o FrFT)	81.83	70.01	3.89	18.17	864.99	28.69
Proposed (w/ FrFT)	82.56	71.51	6.64	32.83	865.50	49.17
Methods	YC-YC Dataset					
	OA (%)	κ (%)	Tr_Time (s)	Te_Time (s)	Parameters (K)	Flops (M)
SSWADA	90.79	87.13	10.77	100.86	33682.87	41.83
FDDAN	90.67	87.10	2.77	84.85	4220.05	37.25
FDGNet	86.04	80.42	8.01	329.28	3000.71	157.17
S^4DL	90.98	87.46	3.58	84.60	1787.35	18.21
ADNet	85.40	79.42	4.04	66.72	2844.45	278.38
DHSNet	87.35	82.30	0.47	53.83	6538.38	25.37
Proposed (w/ CNN)	78.47	70.75	23.25	88.12	656.26	23.87
Proposed (w/ Transformer)	90.19	86.26	23.61	97.20	1055.75	16.47
Proposed (w/o FrFT)	92.13	89.01	22.57	90.30	979.34	34.29
Proposed (w/ FrFT)	92.85	90.00	30.16	135.35	979.85	54.78
Methods	HHK-HHK Dataset					
	OA (%)	κ (%)	Tr_Time (s)	Te_Time (s)	Parameters (K)	Flops (M)
SSWADA	88.17	73.43	8.94	203.51	23987.64	28.19
FDDAN	83.38	68.37	12.40	303.51	3501.27	30.54
FDGNet	83.07	62.64	4.68	473.04	2534.27	150.21
S^4DL	88.97	76.48	2.72	163.41	1699.90	16.64
ADNet	84.05	65.35	2.69	93.73	2303.26	209.90
DHSNet	84.15	68.86	0.45	80.15	6531.25	22.65
Proposed (w/ CNN)	88.25	74.58	11.60	143.99	611.35	21.66
Proposed (w/ Transformer)	86.67	71.43	13.33	175.46	1010.96	14.25
Proposed (w/o FrFT)	88.63	74.17	12.71	168.26	934.42	32.08
Proposed (w/ FrFT)	90.02	78.34	17.89	224.34	934.93	52.57

V. CONCLUSION

In this article, a new HSI-FrFRWKV model has been proposed from the perspective of amplitude–phase joint analysis for cross-scene wetland mapping on HSIs. First, FrFT is innovatively integrated into the RWKV block to design an FrFT-RWKV block. A hierarchical FrFT-RWKV FRNet is further built, followed by a parameter-free FAP loss to supervise its optimization process. Therefore, the spatial–spectral–amplitude–phase features can be extracted in the optimal domain between the space and frequency domains, effectively emphasizing the frequency and structure information of HSIs. Finally, the replay-based self-distillation learning strategy is proposed to improve the training stability of the bi-classifier adversarial network and alleviate its catastrophic forgetting issue for better cross-scene wetland mapping on HSIs. Extensive quantitative and qualitative analysis on a public cross-domain dataset and two wetland datasets demonstrate that the proposed HSI-FrFRWKV model outperforms other competitors.

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